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ROSSMANN STORE SALES FORECASTING

THE PROBLEM & THE DATA

Real-Life Problem Statement

Rossmann, a major European drug-store chain, tasks its store managers with predicting daily sales up to six weeks in advance. However, these manual predictions, based on unique local circumstances, lead to highly varied and unreliable accuracy.

Store sales are influenced by a complex set of factors—including promotions, competition, school and state holidays, seasonality, and locality—which are extremely difficult for managers to model manually and consistently. This project addresses this real-life problem by using machine learning to provide accurate, data-driven sales forecasts, enabling optimized resource allocation.

Dataset & Preprocessing

- Source: The dataset was sourced from a Kaggle competition, providing historical sales data for 1,115 Rossmann stores.
- Attributes: The data included store.csv (store-specific info like 'StoreType', 'CompetitionDistance') and train.csv (historical data like 'Date', 'Sales', 'Promo', 'StateHoliday').
- Preprocessing Steps:
  a. Handling Missing Values: Critical features like 'CompetitionDistance' had missing values, which were handled using mean/median imputation. Stores that were permanently closed (sales=0) were filtered out to prevent model skew.
  b. Encoding: Categorical variables like 'StoreType' and 'StateHoliday' were one-hot encoded for the model.
  c. Scaling: Numerical cols were normalized using Min-Max scaling.

IOT INTEGRATION & DATA PIPELINE

While this project used a static dataset, a real-world MLOps solution would integrate real-time data from IoT-enabled devices:

- Data Ingestion: Point-of-Sale (POS) systems act as IoT endpoints, streaming live transaction data ('Sales', 'Customers') to a central data lake.
- Real-Time Features: IoT customer-counting sensors (e.g., door-mounted beacons) could provide real-time foot traffic, adding a powerful predictor that static data lacks.
- Automation: This live data stream would feed the MLOps pipeline, enabling continuous monitoring and model re-training.

MLOPS & MODEL SELECTION

Comparative Analysis & Model Selection

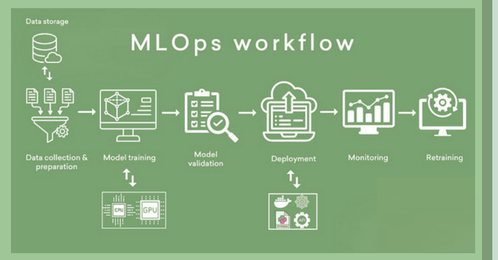
A comparative analysis was performed to select the best regression model. We evaluated a baseline Linear Regression model against a gradient-boosting model (XGBoost).

- Criteria: The primary evaluation metric was Root Mean Square Error (RMSE), as it penalizes large errors heavily, which is crucial for business forecasting.
- Selection: The XGBoost Regressor was chosen as the final model. It dramatically outperformed Linear Regression, achieving an RMSE of 972.66 (compared to 2735.54). This demonstrates its superior ability to capture the complex, non-linear relationships between promotions, holidays, and sales.

Model	Validation RMSE	Train RMSE
XGBoost (100 estimators, 0.7 LR)	**972.66**	952.18
XGBoost (50 estimators, 0.5 LR)	1278.8	1265.21
Linear Regression	2735.54	2726.37

MLOps Workflow for Intelligent Deployment

To deploy this model as an "intelligent solution," a full MLOps workflow is required.This moves beyond a 1-time analysis and creates a system that continuously learns & delivers value.



REAL-LIFE APPLICATIONS & CONCLUSION

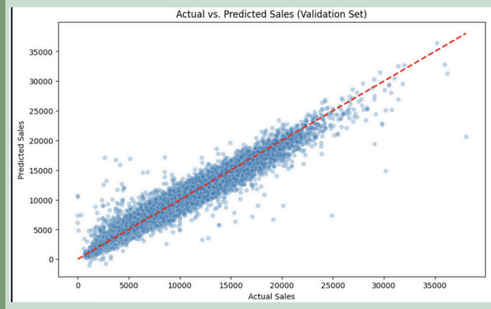
This project successfully demonstrates that a machine learning model (XGBoost) can accurately forecast daily retail sales by modeling complex data. The findings confirm that promotions are the strongest driver of sales, providing a clear, actionable business insight.

This solution provides a framework that directly supports business objectives and aligns with SDG 8 & 9:

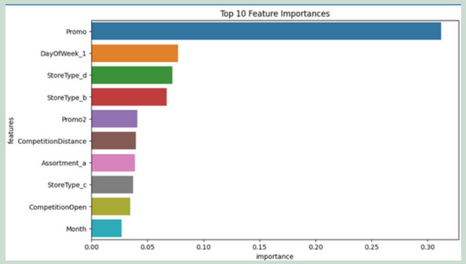
- Staff Scheduling: Managers can use the forecasts to create effective staff schedules, optimizing productivity and supporting SDG 8: Decent Work. & Inventory Management: Reliable forecasts help optimize stock levels, especially for promotions.
- Marketing Strategy: Allows for data-driven decisions on when and where to run promotions to maximize sales and build resilient retail infrastructure (SDG 9).

RESULTS & CONCLUSIONS

Model Performance Evaluation & Results



- Key Insight: The scatter plot shows a tight, strong positive correlation. The model is unbiased, with predictions evenly scattered around the 45-degree line, indicating high accuracy.
- Final Result: The tuned model achieved a Validation RMSE of 972.66 and a Training RMSE of 952.18. The close-in-value scores show that the model is well-generalized and not overfitting.



- Key Insight: This graph clearly shows that 'Promo' (whether a promotion is active) is the single most important predictor of sales.
- Other critical factors include 'CompetitionDistance', 'Store' (Store-specific behavior), and 'DayOfWeek'. This analysis provides actionable insights for business managers.